

- SMBus specification rev 3.0 compatibility:
 - Hardware PEC (packet error checking) generation and verification with ACK control
 - Command and data acknowledge control
 - Address resolution protocol (ARP) support
 - Host and device support
 - SMBus alert
 - Timeouts and idle condition detection
- PMBus rev 1.3 standard compatibility
- Independent clock: a choice of independent clock sources allowing the I²C-bus communication speed to be independent of the PCLK reprogramming
- Wake-up from Stop mode on address match
- Programmable analog and digital noise filters
- 1-byte buffer with DMA capability

Table 8. I2C implementation

I2C features ⁽¹⁾	I2C1
Standard-mode (up to 100 kbit/s)	X
Fast-mode (up to 400 kbit/s)	X
Fast-mode Plus (up to 1 Mbit/s) with extra output drive I/Os	X
Programmable analog and digital noise filters	X
SMBus/PMBus hardware support	X
Independent clock	X
Wake-up from Stop mode on address match	X

1. X: supported

3.18 Universal synchronous/asynchronous receiver transmitter (USART)

The devices embed two universal synchronous/asynchronous receivers/transmitters that communicate at speeds of up to 6 Mbit/s.

They provide hardware management of the CTS, RTS and RS485 DE signals, multiprocessor communication mode, synchronous SPI communication and single-wire half-duplex communication mode. Some can also support SmartCard communication (ISO 7816), IrDA SIR ENDEC, LIN Master/Slave capability and auto baud rate feature, and have a clock domain independent of the CPU clock, which allows them to wake up the MCU from Stop mode. The wake-up events from Stop mode are programmable and can be:

- start bit detection
- any received data frame
- a specific programmed data frame

All USART interfaces can be served by the DMA controller.